CLINICAL PROGRAMMING BOOTCAMP • MODULE 09

Derivations

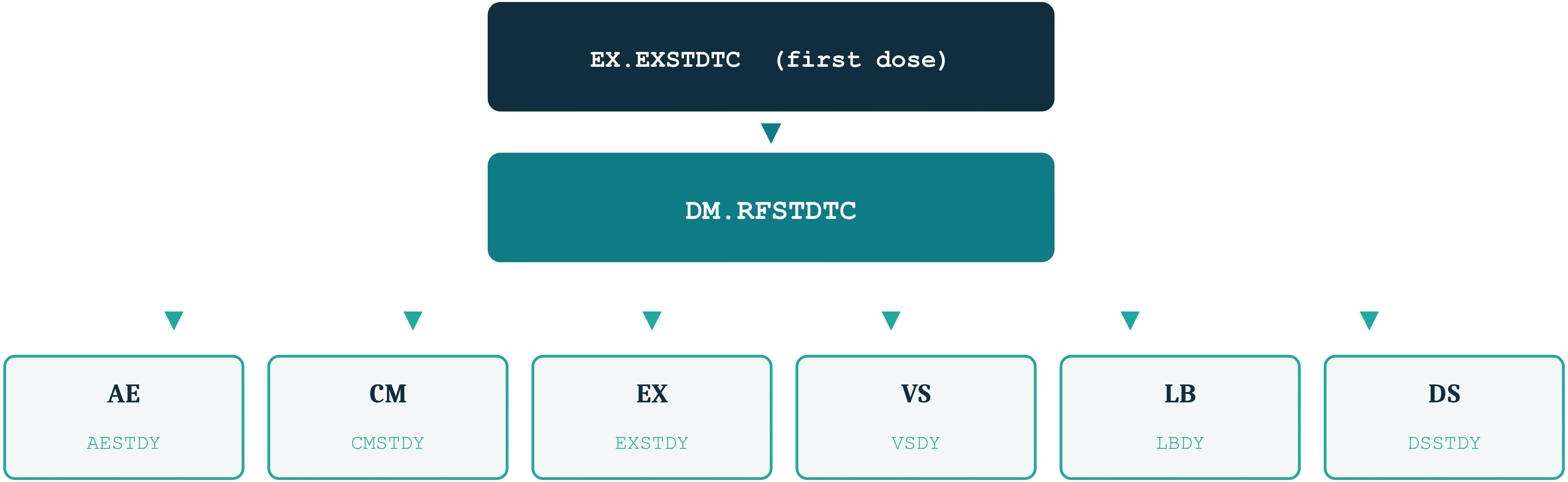
Study day, --SEQ, and the shape of study time

You have derived --DY and --SEQ in six domains already. This module turns six separate habits into one rule you can apply anywhere.

Hands-on: Notebook 11 · Deriving --DY and --SEQ (SAS)

EVERYTHING HANGS OFF ONE DATE

RFSTDTC — the reference start date



One date, 226 derived values. In ABC-01 there are 226 --DY values across six domains, and every one is measured from that subject's own RFSTDTC. An EX date that is wrong by a day shifts the entire study timeline — and shifts it CONSISTENTLY, so no within-domain check will ever catch it.

THE RULE

One formula, two branches, no Day 0

The only study-day rule you need

```
if   --DTC >= RFSTDTC   then   --DY = (--DTC - RFSTDTC) + 1;     /* on or after */
else                           --DY = (--DTC - RFSTDTC);         /* before      */
```

Why the asymmetry? Because Day 0 does not exist.

−3	−2	−1	1	2	3
		day before	FIRST DOSE		

There is no box between −1 and 1.

Proof from your own data

Across all 226 --DY values in ABC-01 — including 61 negative ones — there is not a single zero. That is not luck: the "+ 1" on one branch only is what enforces it. If a 0 ever appears in your output, one branch is missing its + 1.

NEGATIVE IS NOT AN ERROR

It depends entirely on what the domain collects

Domain	Variable	Range in ABC-01	Negative	Why
EX	EXSTDY	1 … 1	0	dosing defines Day 1 — it cannot precede itself
LB	LBDY	1 … 29	0	no screening blood draw in this protocol
AE	AESTDY	−5 … 20	1	collected from consent — one screening event
CM	CMSTDY	−96 … 15	4	prior medications go back months
DS	DSSTDY	−11 … 28	8	informed consent precedes dosing for everyone
VS	VSDY	−11 … 29	48	screening visit vitals for all 8 subjects

Expect negatives

CM · prior medications, often months before
VS · screening assessments
DS · the informed-consent milestone
AE · screening-period events

Question a negative

EX · a dose before first dose is impossible
LB · only if the protocol has no screening draw

A rule that flags EVERY negative --DY as an error would fire on 61 perfectly valid records here.

--SEQ: making every record addressable

What it is for

USUBJID identifies the subject. --SEQ identifies WHICH RECORD within that subject and domain. Together they are the key that lets anything else — SUPPQUAL, RELREC, a query, a reviewer's note — point at one specific row.

Domain	Subjects	Max --SEQ	Rows per subject	What --SEQ counts
EX	8	1	1	one dosing period each
AE	6	2	1–2	adverse events
CM	7	2	1–2	medications
DS	8	3	3	disposition milestones
LB	4	12	12	6 tests × 2 visits
VS	8	16	16	6 tests at screening + 5 × 2 later visits

--SEQ restarts at 1 for every subject. It is not a row number for the dataset. Note only 4 subjects appear in LB and 6 in AE — a subject with no records in a domain simply is not there.

THE PART PEOPLE GET WRONG

--SEQ is only as reproducible as your sort

The pattern used in every domain

```
proc sort data = ae_work;
    by usubjid aestdtc aeterm;      /* <-- the TIEBREAKER is the point */
run;

data ae;
    set ae_work;
    by usubjid;
    retain aeseq;
    if first.usubjid then aeseq = 1;   /* restart for each subject */
    else                  aeseq + 1;
run;
```

⚠ Sorting by date alone

Two events on the same day have no defined order, so SAS may emit them either way. Re-run the program and AESEQ 1 and 2 can swap — and every SUPPQUAL row that pointed at AESEQ 1 now points at the other event.

Deterministic sort

Add a tiebreaker that cannot tie: date THEN term. Now the same input always gives the same --SEQ, on any machine, in any SAS version — which is what makes a re-run comparable to the original.

EPOCH — which phase of the trial a record belongs to

--DY tells you WHEN relative to first dose. EPOCH tells you WHICH PART of the trial that was.

SCREENING consent → first dose	`--DY < 1`	AE sore throat (−5) · VS screening (−11…−9)
TREATMENT first dose → last dose	`1 ≤ --DY ≤ end of dosing`	most of the study's records
FOLLOW-UP after last dose	`--DY > end of dosing`	the Week 4 visit for some subjects

ABC-01 does not populate EPOCH. It is Permissible, and this study is small enough not to need it. Know that it exists and that it is derived from the same reference dates — you will meet it the moment you work on a study with multiple treatment periods.

BEFORE NOTEBOOK 11

Five ways to get timing wrong

1	**A study day of 0** One branch is missing its + 1. There is no Day 0.	*→ check: no --DY equals 0*
2	**Treating negatives as errors** 61 of the 226 --DY values here are legitimately negative.	*→ ask what the CRF collects*
3	**--DY from a study-wide date** Each subject has their own RFSTDTC. Screening ranges −11 to −9.	*→ always per subject*
4	**--SEQ numbered across the dataset** It restarts at 1 for every subject.	*→ by usubjid; if first.usubjid*
5	**Sorting without a tiebreaker** Same-day records can swap between runs, breaking every SUPP link.	*→ sort by date THEN term*

Next: Notebook 11 · Deriving --DY and --SEQ → Day 9 · Validation